

Notes: Guiso and Zaccaria (JFE, 2023)

From Patriarchy to Partnership: Gender Equality and Household Finance

Contents

1	Big picture	2
2	Incremental contribution of the paper	2
3	Data and measurement	3
3.1	Main data source: SHIW	3
3.2	Household headship	4
3.3	Financial outcomes	4
3.4	Comparative advantage controls	4
4	Conceptual framework	5
4.1	Two-stage household financial decision making	5
4.2	Headship with social conformity	5
4.3	Empirical analogue of social norms	6
4.4	Selection and collaboration channels	6
5	Empirical specifications	6
5.1	Measuring gender norms	6
5.2	Financial participation regressions	7
5.3	Return regressions	7
5.4	Mechanism and heterogeneity tests	7
5.5	Pension reform design	8
6	Identification and empirical design	8
6.1	What the design can identify	8
6.2	Main identifying assumption for Equality	8
6.3	Why non-couple households are a placebo	8
6.4	Potential limitations	9
7	Tables and figures	9
7.1	Figure 1: Female headship in two-spouse households; Figure 2: Female headship cohort and region effects	9
7.2	Figure 3: Sample size by household type; Figure 4: Female headship by cohort and geographical area	10

7.3	Table 1: Household characteristics; Table 2: Investment in financial markets	11
7.4	Figure 5: Participation in financial markets by cohort and region; Table 3: Determinants of female headship	11
7.5	Figure 6: Social norms estimates; Figure 7: Political views, religiosity, and reproductive health services	12
7.6	Table 4: Spousal equality and investment in financial assets	13
7.7	Table 5: Spousal equality, asset allocation, and diversification; Table 6: Spousal equality and financial returns	14
7.8	Table 7: Role of collaboration; scope and incentives	15
7.9	Table 8: Non-couple households	16
7.10	Figure 8: Pension reform; Table 9: Treated versus control households	17
7.11	Table 10: Effects of the pension reform	17
8	Short summary	18
9	Conclusion	19
9.1	Core conclusion	19
9.2	Economic intuition	19
9.3	Incremental contribution revisited	19
9.4	Implications for household finance	19
9.5	Implications for gender economics	19
9.6	Limitations	19
9.7	Takeaway	20

1 Big picture

- **Question.** Do gender norms inside the household affect household financial decisions and investment performance? The paper asks whether the traditional rule that the husband should manage money creates economic costs.
- **Setting.** The paper studies Italian households using the Bank of Italy Survey of Households Income and Wealth (SHIW), focusing on two-spouse households from 1991 to 2014.
- **Key institutional fact.** The person reported as the household head is the person in charge of, or most informed about, household economic and financial management. This makes headship a direct proxy for financial decision power inside the family.
- **Main empirical fact.** Female financial headship among Italian two-spouse households rises dramatically, from about 1 percent in the early 1990s to more than 35 percent in 2014.
- **Core measurement idea.** The authors estimate a model-driven measure of local gender norms by explaining female headship with spousal comparative advantage controls and then using cohort-region fixed effects as the residual social-norm component.
- **Main explanatory variable.** The estimated cohort-region effect is called *Equality*. Higher values indicate more egalitarian gender norms in the relevant reference group.
- **Main result.** More egalitarian norms increase participation in financial markets, holdings of stocks and bonds, pension fund participation, asset-class diversification, and measured investment returns.
- **Mechanism.** The evidence is consistent with two channels: better *selection* of the spouse with higher financial-management ability and more *collaboration* between spouses when norms become more egalitarian.
- **Cultural-change result.** Patriarchal norms began receding sharply in the early 1990s. The authors connect this shift to the 1992 Italian pension reform, which lowered expected public pension wealth and made inefficient household financial management more costly.
- **Economic takeaway.** Gender norms are not only social or identity constraints. They can distort who makes financial decisions, reduce market participation, lower diversification, and reduce household returns.

2 Incremental contribution of the paper

- **From gender gaps to intra-household finance.** Much of the gender-and-finance literature documents gender differences in risk taking, overconfidence, literacy, or participation. This paper studies a different mechanism: how gender norms allocate financial authority inside the household.
- **Behavior-based measure of norms.** Instead of using survey opinions from sources such as the World Values Survey, the paper constructs norms from actual household behavior - the gender of the reported financial head.

- **Cohort-region identification of social norms.** The paper exploits large variation across birth cohorts and Italian regions in female headship after controlling for relative education, income, hours worked, age, and property ownership.
- **Model-guided interpretation.** A simple social-conformism model turns the cohort-region headship fixed effect into a measure of gender norms and clarifies how those norms affect financial outcomes through selection and collaboration.
- **Household-finance outcome contribution.** The paper links norms to a broad set of outcomes: financial market participation, asset-class choices, diversification, leverage and real-asset allocation (in the appendix), and returns.
- **Mechanism tests.** The paper goes beyond a baseline association by testing predictable heterogeneity: effects are stronger when spouses can share information or attention costs, weaker when both spouses have very low or very high education, weaker among foreign-born households, and weaker among wealthier households.
- **Placebo strategy.** The authors show that the equality measure has no comparable effect on non-couple households or single women, reducing concerns that the measure simply captures broad regional liberalism, trust, secularization, or female labor-market development.
- **Cultural evolution contribution.** By studying the 1992 pension reform, the paper offers evidence on why social norms change: a large economic shock can make the cost of obeying old norms exceed the comfort of conforming.
- **Finance and welfare contribution.** The paper quantifies that one standard deviation in equality raises financial-market participation by about 3 percentage points and improves returns by economically meaningful magnitudes.

3 Data and measurement

3.1 Main data source: SHIW

- The paper uses the Italian Survey of Households Income and Wealth (SHIW), run by the Bank of Italy.
- The sample period for the main analysis is 1991–2014, because the key household-finance and headship variables are available consistently over this period.
- SHIW contains roughly 8,000 households per wave and covers all 20 Italian regions.
- The survey contains demographics, education, income, employment, savings, assets, debt, and portfolio holdings.
- The analysis focuses on two-spouse households, married or cohabiting, because these households face a within-couple allocation problem over who manages financial decisions.

Interpretation: The data are well suited to the paper’s question because they contain both the identity of the household financial head and detailed financial portfolio information.

3.2 Household headship

- In SHIW, household members identify the household head at the beginning of the interview.
- The head is defined as the person in charge of, or most informed about, household economic and financial management.
- From 1991 to 2006, the wording emphasizes being in charge of economic and financial management. In later waves, the wording includes being more informed, and from 2012 the label changes to reference person.
- The authors find no discontinuity around wording changes, suggesting that the meaning is stable over time.
- Female headship is interpreted as female financial-management authority, not simply demographic seniority.

Interpretation: The headship variable is central because it transforms the abstract idea of gender norms into an observable household governance choice.

3.3 Financial outcomes

The paper studies several household financial outcomes:

- **Participation in financial markets:** an indicator equal to one if the household holds any financial asset beyond bank deposits, including government bonds, other fixed income, stocks, funds, or other securities.
- **Asset allocation:** indicators for specific asset classes such as stocks, government bonds, other bonds, other securities, and pension funds.
- **Diversification:** the number of different financial asset classes held.
- **Financial return:** financial income divided by financial assets.
- **Total return:** capital income divided by total assets.
- **Total net return:** net capital income divided by net worth.

Interpretation: Participation and portfolio choice capture decision quality on the extensive and intensive margins, while returns give a direct but noisier welfare-relevant outcome.

3.4 Comparative advantage controls

To separate social norms from efficient specialization, the authors control for observable wife-versus-husband comparative advantage:

- female-to-male income ratio;
- female share of household real-estate ownership;
- female-to-male education ratio;

- female-to-male age ratio;
- whether the female spouse is a housewife;
- female and male hours worked;
- occupation fixed effects for both spouses.

Interpretation: These variables are meant to absorb the Becker-style efficient allocation channel. The residual cohort-region variation in female headship is then interpreted as social norms.

4 Conceptual framework

4.1 Two-stage household financial decision making

The model assumes a two-stage process.

1. **Headship stage.** The household chooses whether the husband or wife will manage household wealth.
2. **Investment stage.** The chosen manager implements the household financial decision.

Let $G = 1$ denote the female spouse and $G = 0$ the male spouse. Each spouse has financial ability $X_{i,G}$. The household can keep savings in a safe deposit D with return r , or invest in a sophisticated risky asset S with expected return advantage that depends on the manager's skill.

If the manager's skill and investment opportunity are good enough, the risky asset yields $R > r$; otherwise it yields zero. Under the paper's simple setup, the expected return is

$$Y_i = X_{i,G}\lambda + r,$$

where $\lambda = R - r$ is the premium from investing in the sophisticated asset.

Interpretation: The value of choosing the right household financial manager is higher when financial markets offer opportunities that require skill to evaluate.

4.2 Headship with social conformity

At $t = 0$, household i in community z assigns headship to spouse G to maximize

$$\alpha U_{i,G} - \beta_z (G - \bar{G}_z)^2.$$

The first term captures intrinsic utility from choosing the more capable or lower-cost manager. The second term captures the social cost of deviating from the prevailing local gender norm $\bar{G}_z$.

If $\bar{G}_z = 0$, the community is patriarchal and the social norm favors male headship. If $\bar{G}_z = 1/2$, the norm is gender neutral.

The probability of female headship can be written as

$$\Pr(G_i = 1) = \Phi(\Delta X_i \lambda + C_z),$$

where C_z is the community-specific gender norm component. A larger C_z means more egalitarian norms.

Interpretation: Holding relative spouse characteristics fixed, a household in a more egalitarian community is more likely to assign financial headship to the wife.

4.3 Empirical analogue of social norms

Because the econometrician observes only part of financial ability, the paper writes

$$X_{i,G} = A_{i,G} + Z_{i,G},$$

where $A_{i,G}$ is observable ability and $Z_{i,G}$ is unobservable ability.

The empirical female-headship equation becomes

$$\Pr(G_i = 1) = \Phi(\Delta A_i \lambda + \Delta Z_i \lambda + C_z).$$

The empirical strategy estimates C_z as a cohort-region fixed effect after controlling for observed comparative advantage ΔA_i .

Interpretation: The empirical equality measure is not simply female headship. It is female headship unexplained by observable spousal characteristics.

4.4 Selection and collaboration channels

The model implies two channels.

Selection channel. In patriarchal communities, male spouses can become financial heads even if they are not the best financial decision makers. As norms become more egalitarian, selection into headship becomes more skill based.

Collaboration channel. Egalitarian norms can also make spouses collaborate more, share information, divide attention costs, and exploit complementary knowledge. This channel improves outcomes for both male- and female-headed households.

Interpretation: The selection channel predicts weaker equality effects among female-headed households because the women who become heads in patriarchal communities are already highly selected. The collaboration channel predicts positive effects for households more generally.

5 Empirical specifications

5.1 Measuring gender norms

The baseline headship model is

$$G_{i,c,r} = \alpha + \Delta A_i \lambda_1 + \Psi_i \gamma + C_{c,r} + \varepsilon_{i,c,r},$$

where $G_{i,c,r}$ equals one if household i in cohort c and region r is female-headed.

The estimated cohort-region fixed effect $\hat{C}_{c,r}$ is the paper's measure of *Equality*. Higher values mean less patriarchal and more gender-neutral local norms.

Interpretation: The construction of Equality is similar to using group fixed effects to measure culture, but the group-level object comes from actual household behavior rather than survey answers.

5.2 Financial participation regressions

The main second-stage regression relates financial outcomes to the generated Equality measure:

$$Y_{i,c,r,t} = \theta \widehat{C}_{c,r} + X'_{i,t} \Gamma + \delta_{r,t} + \omega_{occ} + v_{sector} + u_{i,c,r,t},$$

where Y is a financial decision or return outcome. Controls include household demographics, income and wealth deciles, head age and education, comparative advantage measures, region-by-year fixed effects, occupation fixed effects, and sector fixed effects.

Because Equality is generated in the first stage, the authors use a two-stage bootstrap procedure for standard errors.

Interpretation: The second stage asks whether households from more egalitarian cohort-region groups make better financial decisions, conditional on rich observables and fixed effects.

5.3 Return regressions

The return regressions use financial return, total return, and total net return as dependent variables. The authors estimate them both in pooled cross sections and in the panel sample, where household returns are averaged across survey years.

Interpretation: Return regressions are more directly welfare related but also noisier because SHIW is not a high-frequency portfolio-performance dataset.

5.4 Mechanism and heterogeneity tests

The paper studies whether equality effects are stronger where collaboration should matter more:

- whether the non-head spouse works in finance;
- whether spouses work in the same sector;
- relative hours worked by head and non-head spouse;
- whether the couple has children.

It also studies settings where norms should matter less:

- both spouses have very low or very high education;
- one or both spouses are foreign born;
- households are wealthy.

Interpretation: These tests are important because they check whether the baseline effect behaves as a gender-norm mechanism should behave.

5.5 Pension reform design

To study why norms changed, the paper analyzes the 1992 Italian pension reform. The reform reduced expected public pension benefits, especially for workers with less than 15 years of tenure at the end of 1992.

The difference-in-differences specification uses households in 1989, 1991, 1993, and 1995. A household is treated if at least one member was affected by the reform. The key coefficient is on $Post \times Treated$.

Interpretation: The reform increased the cost of poor private financial decisions. If patriarchal norms misallocate financial decision power, the reform should make families more likely to abandon male-default headship.

6 Identification and empirical design

6.1 What the design can identify

- The paper identifies whether cohort-region differences in estimated gender norms predict household financial decisions after controlling for observed spousal characteristics and many fixed effects.
- The evidence is not a randomized experiment in the baseline Equality regressions. It is a model-guided observational design with extensive robustness and placebo tests.
- The pension reform analysis is closer to a quasi-experimental design, using differential exposure to a national policy shock.

Interpretation: The paper's strength is that it combines a structural measurement idea, rich controls, heterogeneity tests, placebo tests, and a policy-shock analysis.

6.2 Main identifying assumption for Equality

The key assumption is that unobserved female-versus-male financial ability differences are not systematically correlated with cohort-region norms after controlling for observable comparative advantage.

This could fail if women in patriarchal areas have lower unobserved ability, or if high-ability women opt out of marriage in patriarchal communities. The paper addresses this concern with tests on non-couple households and single women.

Interpretation: This assumption is central because Equality is inferred from residual headship differences. The placebo evidence is designed to show that these residuals are not simply broad female ability or labor-market differences.

6.3 Why non-couple households are a placebo

If Equality were only capturing secularization, trust, political progressivism, or female labor-market development, it should also predict financial behavior among singles, including single women. The

paper finds no such effect.

Interpretation: This placebo supports the interpretation that Equality operates through intra-household couple dynamics, not general regional culture alone.

6.4 Potential limitations

- The Equality measure is generated from survey-reported headship, not observed transaction-level decision authority.
- The baseline relation between Equality and finance may still contain omitted variables that pass through couple-specific channels.
- The return measures are noisy because the data are survey based and not a complete administrative record of portfolio returns.
- Italy’s institutional and cultural environment may limit external validity to countries with different household structures or financial systems.

7 Tables and figures

7.1 Figure 1: Female headship in two-spouse households; Figure 2: Female headship cohort and region effects

Read:

- Figure 1 shows a dramatic rise in female headship, especially after the early 1990s.
- The share of two-spouse households headed by women rises from about 1 percent to more than 35 percent by 2014.
- Figure 2 decomposes headship into cohort and region effects after controlling for spouse characteristics.
- The cohort effect is strongly increasing: younger birth cohorts are much more likely to have female headship.
- The region effect is heterogeneous and does not simply follow the North-South income gradient.

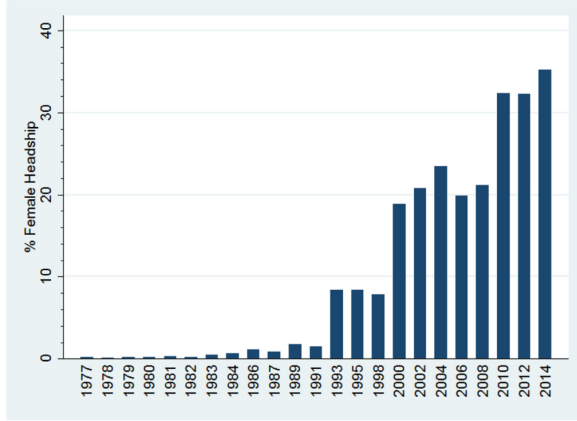
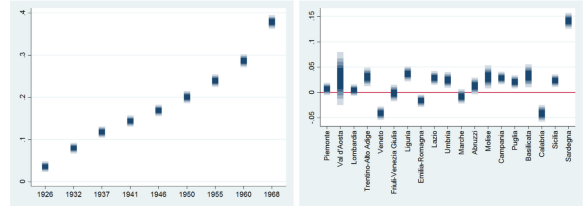


Fig. 1. Female headship in two-spouse households.

(a) Figure 1: Female headship over time



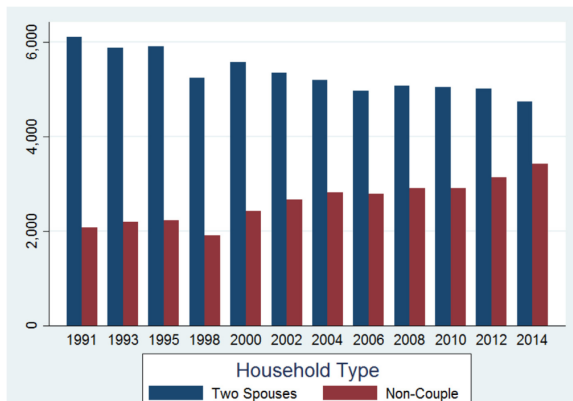
(b) Figure 2: cohort and region effects

Economic message: The raw time series shows that Italian households experienced a large governance transition. The cohort-region pattern motivates the paper's treatment of social norms as local and generational rather than purely national.

7.2 Figure 3: Sample size by household type; Figure 4: Female headship by cohort and geographical area

Read:

- Figure 3 shows that two-spouse households remain the main household type but decline over time, while non-couple households grow.
- Figure 4 shows that female headship rises in all cohorts and areas, but at different speeds.
- Younger cohorts move toward female headship much faster.
- The area plot shows that the transformation is nationwide but heterogeneous across Italy.



(a) Figure 3: household type composition

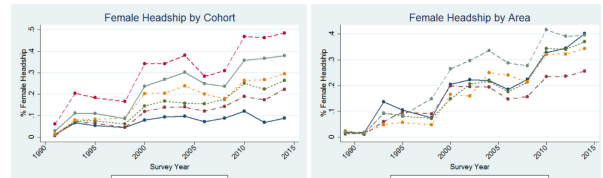


Fig. 3. Sample Size by Household Type and Survey Year. This figure displays the number of households in each survey year by household type. Two spouses households consist of two adults of opposite sex married or cohabiting plus other members (e.g., children or parents). Non-couple households are households that do not include an adult couple, e.g., single-parent households.

(b) Figure 4: female headship by cohort and area

Economic message: The increasing female-headship trend is not just a change in household

composition. It appears inside two-spouse households and varies predictably by cohort and region.

7.3 Table 1: Household characteristics; Table 2: Investment in financial markets

Read:

- Table 1 shows that female-headed households differ from male-headed households in economically meaningful ways.
- Female heads have higher relative income, education, and sole real-estate ownership than women in male-headed households.
- Table 2 shows that 27 percent of two-spouse households participate in at least one financial asset class beyond bank deposits.
- Stock-market participation is low at 7 percent, while government bond participation is 15 percent and fund participation is 9 percent.
- Average financial return is about 3 percent, while total net return is much lower.

Table 1
Household Characteristics.
This table shows summary statistics of two-spouse households characteristics in the full sample and by gender of household head. "NorthWest" indicates the fraction of households living in the regions of Liguria, Piemonte, Val d'Aosta, and Lombardia; "NorthEast" the fraction of households living in the regions of Veneto, Trentino-Alto Adige, Friuli Venezia Giulia, and Emilia Romagna; "Center" the fraction of households living in the regions of Toscana, Umbria, Lazio, and Marche; "South" the fraction of households living in the regions of Abruzzi, Molise, Campania, Puglia, Basilicata, and Calabria; "Islands" indicates the fraction of households living in the regions of Sicilia and Sardegna.

	Mean	Median	Min	Max	Mean:M Head	Mean:F Head	Δt -stat	Obs
Demographic and other controls								
# Adults	2.07	2.00	2.00	9.00	2.07	2.09	-4.14	64,085
# Children	1.16	1.00	0.00	7.00	1.16	1.14	2.50	64,085
Avg Age	52.94	52.50	17.50	96.50	53.09	52.29	5.69	64,085
Retired	0.18	0.00	0.00	1.00	0.19	0.17	4.36	64,085
Avg Education	3.05	3.00	1.00	6.00	3.04	3.11	-7.53	64,085
Home Owner	0.72	1.00	0.00	1.00	0.72	0.70	4.35	64,085
Cohabiting Couple	0.02	0.00	0.00	1.00	0.02	0.05	-19.08	64,085
Comparative advantage measures								
Education FIM ratio	1.00	1.00	0.17	4.00	0.99	1.06	-24.05	64,042
Income FIM ratio	0.41	0.28	0.00	2.96	0.31	0.89	-115.78	63,529
RE Female Ownership	0.09	0.00	0.00	1.00	0.08	0.16	-33.01	64,085
Age FIM ratio	0.93	0.94	0.15	3.05	0.93	0.94	-5.22	64,042
Female is Housewife	0.43	0.00	0.00	1.00	0.44	0.39	9.90	64,085
Hours Worked F	10.34	0.00	0.00	121.85	10.00	11.80	-11.16	63,860
Hours Worked M	22.22	33.23	0.00	138.46	22.08	22.85	-3.76	63,959

(a) Table 1: household characteristics

Table 2
Two-spouse household: investment in financial markets.
The upper panel of this table shows the proportion of two-spouse households that participate in financial markets in general and in specific asset classes. The lower panel provides summary statistics for the number of different asset classes held, the ratio of financial income over financial assets, and the ratio of net capital income over total assets.

	Mean	p25	Median	p75	StDev	Obs
Participation						
Any Fin Asset Class	0.27	0.00	0.00	1.00	0.45	64,085
Gov Bonds	0.15	0.00	0.00	0.00	0.36	64,085
Other Fixed Income	0.07	0.00	0.00	0.00	0.25	64,085
Stocks	0.07	0.00	0.00	0.00	0.26	64,085
Funds	0.09	0.00	0.00	0.00	0.29	64,085
Other Securities	0.00	0.00	0.00	0.00	0.02	64,085
# Fin Asset Classes	1.44	1.00	1.00	2.00	0.76	17,524
Fin Income/Fin Assets	3.08	1.19	2.76	4.40	2.25	55,115
Net Capital Income/Total Assets	0.53	0.01	0.14	0.75	1.49	63,166

(b) Table 2: financial market participation and returns

Economic message: Female headship is associated with observed comparative advantage, confirming the need to control for spousal characteristics. The financial outcome variables show substantial room for household-finance inefficiency.

7.4 Figure 5: Participation in financial markets by cohort and region; Table 3: Determinants of female headship

Read:

- Figure 5 shows that participation is higher in older cohorts and in richer Northern regions.
- This pattern differs from the female-headship pattern, which helps separate gender norms from ordinary financial development.
- Table 3 estimates female headship as a function of comparative advantage and social-group effects.

- Income ratio, female real-estate ownership, and education ratio all positively predict female headship.
- Cohort-region fixed effects raise adjusted R^2 substantially, from about 0.29 to about 0.38.



(a) Figure 5: financial participation by cohort and region

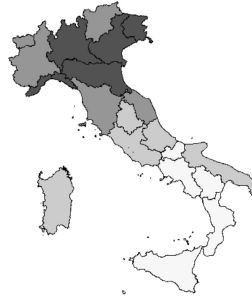


Table 3

Determinants of female headship.

This table reports estimates from the following regression: $FHead_{i,t} = \Delta A_i \beta + \Psi_i \gamma + C_{i,t} + \epsilon_{i,t}$, where $FHead$ is a dummy variable that takes the value 1 if the head of household i is female, ΔA_i is a vector of female-male differences in observable characteristics, Ψ_i is a vector of household-level controls, c indicates cohort, and t indicates region. Standard errors (in brackets) are clustered at the region level. *, **, and *** indicate statistical significance at the 10%, 5%, and 1%, respectively.

	(1)	(2)	(3)	(4)
Income FtM ratio	0.420*** (0.015)	0.469*** (0.015)	0.465*** (0.015)	0.465*** (0.014)
RE Female Ownership	0.094*** (0.010)	0.111*** (0.013)	0.110*** (0.013)	0.110*** (0.013)
Education FtM ratio	0.016** (0.007)	0.035** (0.013)	0.037** (0.013)	0.037** (0.013)
Age FtM ratio	-0.195*** (0.035)	0.015 (0.030)	-0.011 (0.034)	-0.010 (0.034)
Cohabiting Couple	0.071*** (0.014)	0.093*** (0.015)	0.092*** (0.014)	0.092*** (0.014)
#Adults	0.019*** (0.005)	0.013*** (0.005)	0.015*** (0.005)	0.015*** (0.005)
#Children	-0.006** (0.003)	-0.003 (0.003)	-0.005 (0.003)	-0.005 (0.003)
Avg. Education	-0.025*** (0.003)	0.013*** (0.004)	0.009* (0.005)	0.009* (0.005)
Avg. Age	0.012*** (0.001)	0.001*** (0.000)	0.002*** (0.000)	0.002*** (0.000)
Hours Worked F	-0.004*** (0.000)	-0.005*** (0.000)	-0.005*** (0.000)	-0.005*** (0.000)
Hours Worked M	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)
Income Dec.	0.002* (0.001)	0.002 (0.001)	0.003* (0.002)	0.003 (0.002)
Wealth Dec.	-0.003** (0.001)	-0.007*** (0.001)	-0.006*** (0.001)	-0.006*** (0.001)
Commerce&Service			-0.014 (0.152)	-0.000 (0.184)
Income Single F			-0.021*** (0.007)	-0.021*** (0.007)
Gender Balance			-0.701* (0.387)	-0.755* (0.429)
Share in Large City				-0.039 (0.124)
Share Women in Finance				0.145 (0.388)
Occupation M and F FE	Yes	Yes	Yes	Yes
CohortXRegion FE	Yes	No	No	No
N	63,238	63,238	61,540	61,540
adj. R ²	0.381	0.287	0.291	0.291

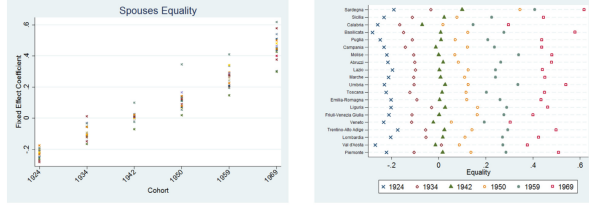
(b) Table 3: determinants of female headship

Economic message: Spousal comparative advantage matters, but it does not fully explain headship. The remaining cohort-region component is the empirical foundation for the Equality measure.

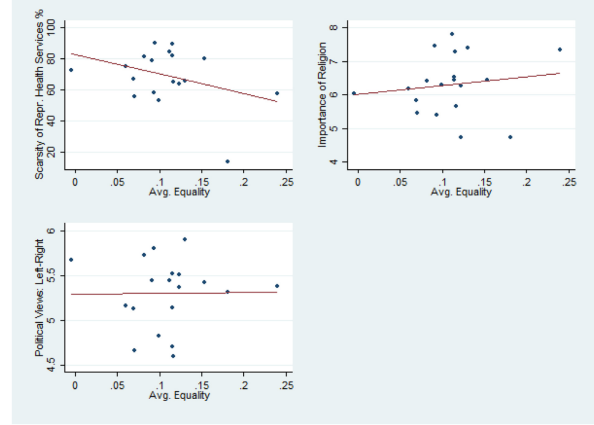
7.5 Figure 6: Social norms estimates; Figure 7: Political views, religiosity, and reproductive health services

Read:

- Figure 6 plots the estimated cohort-region Equality measure.
- Equality increases across cohorts in almost every region, but levels differ substantially across regions.
- Figure 7 compares regional average Equality with other cultural variables.
- Equality is not visibly correlated with left-right politics or religiosity.
- The reproductive-health-services relation is mostly driven by one outlier region.



(a) Figure 6: estimated Equality by cohort and region



(b) Figure 7: cultural correlates of Equality

Economic message: Equality is not just a relabeling of regional political ideology or religiosity. This strengthens the interpretation that the measure captures gender norms in household governance.

7.6 Table 4: Spousal equality and investment in financial assets

Read:

- Table 4 is the first main outcome table.
- Equality has a positive and significant effect on participation in financial assets.
- One standard deviation increase in Equality raises participation by about 3 percentage points, around 10 percent of the sample mean.
- The effect is similar when restricting the sample to heads younger than 65.
- The interaction of Equality with female headship is negative, consistent with the selection channel.

Table 4

Spousal equality and investment in financial assets.

This table shows the coefficient estimates for a linear regression of an indicator variable that takes the value 1 if the household holds wealth in at least one financial asset other than bank deposits. In column (2), we restrict the sample to households with head younger than 65. *Female Head* takes the value 1 if the household head is female. Other Controls include *Income FtM ratio*, *RE Female Ownership*, *Education FtM ratio*, *Age FtM ratio*, *Cohabiting Couple*. Standard errors (in parentheses) are computed with a 2-stages bootstrapping procedure for generated variables. *, **, and *** indicate statistical significance at the 10%, 5%, and 1%, respectively.

	(1)	(2)	(3)
Equality	0.126*** (0.026)	0.133*** (0.032)	0.135*** (0.027)
Equality X Female Head			-0.033* (0.019)
Female Head			0.004 (0.007)
#Adults	-0.040*** (0.004)	-0.040*** (0.005)	-0.040*** (0.004)
#Children	-0.023*** (0.002)	-0.022*** (0.002)	-0.023*** (0.002)
Education	0.042*** (0.002)	0.040*** (0.003)	0.042*** (0.002)
Age	0.009*** (0.001)	0.012*** (0.002)	0.009*** (0.001)
Age ²	-0.0001*** (0.0000)	-0.0001*** (0.0000)	-0.0001*** (0.0000)
Wealth dec.	0.046*** (0.001)	0.046*** (0.001)	0.046*** (0.001)
Income dec.	0.032*** (0.001)	0.030*** (0.001)	0.032*** (0.001)
Home Owner	-0.141*** (0.005)	-0.138*** (0.005)	-0.141*** (0.005)
Hours Worked	-0.0014*** (0.0002)	-0.0013*** (0.0002)	-0.0014*** (0.0002)
Other Controls	Yes	Yes	Yes
Region#Year FE	Yes	Yes	Yes
Occupation M and F FE	Yes	Yes	Yes
Sector HH FE	Yes	Yes	Yes
adj. R ²	0.275	0.271	0.275
Observations	63,457	47,268	63,457

Table 4: Equality and financial-asset participation

Economic message: More egalitarian household norms increase financial-market participation. The muted effect for female-headed households is consistent with high selection of female heads in patriarchal communities.

7.7 Table 5: Spousal equality, asset allocation, and diversification; Table 6: Spousal equality and financial returns

Read:

- Table 5 shows that Equality increases holdings of stocks, government bonds, other bonds, other

securities, and pension funds.

- It also increases the number of asset classes held, implying better diversification.
- Table 6 shows that Equality increases financial return, total return, and total net return.
- The return results hold in the full sample and in the panel sample.
- The authors translate the return effects into meaningful long-run wealth differences across high- and low-Equality groups.

Table 5
Spousal equality, asset allocation and diversification.
This table shows the coefficient estimates for a linear regression of an indicator variable that takes the value 1 if the household holds wealth in stocks (column (1)), government bonds (column (2)), other bonds (column (3)), assets other than stocks, bonds, or investment funds (column (4)), or pension funds (column (5)). Column (6) shows coefficient estimates for a Poisson regression of the number of different asset classes held by the household. Other Controls include *Income FIM ratio*, *RE Female Ownership*, *Education FIM ratio*, *Age FIM ratio*, *Cohabiting Couple*. The sample consists of all couple-households. Standard errors (in parentheses) are computed with a 2-stages bootstrapping procedure for generated variables. *, **, and *** indicate statistical significance at the 10%, 5%, and 1%, respectively.

	(1) Stocks	(2) Gov. bonds	(3) Other bonds	(4) Other securities	(5) Pension funds	(6) #Asset classes
Equality	0.049*** (0.016)	0.060*** (0.021)	0.044*** (0.016)	0.004** (0.002)	0.139*** (0.022)	0.886*** (0.135)
#Adults	-0.012*** (0.002)	-0.019*** (0.004)	-0.008*** (0.003)	-0.000 (0.000)	-0.012*** (0.003)	-0.176*** (0.022)
#Children	-0.004*** (0.001)	-0.014*** (0.001)	-0.007*** (0.001)	-0.0003*** (0.0001)	0.007*** (0.001)	-0.113*** (0.008)
Education	0.024*** (0.001)	0.018*** (0.002)	0.016*** (0.001)	0.000 (0.000)	-0.002 (0.002)	0.164*** (0.008)
Age	0.003*** (0.001)	0.004*** (0.001)	0.003*** (0.001)	0.0002*** (0.0001)	0.005*** (0.001)	0.044*** (0.005)
Age ²	-0.0000*** (0.0000)	-0.0000*** (0.0000)	-0.0000*** (0.0000)	-0.0000*** (0.0000)	-0.0000*** (0.0000)	-0.000*** (0.000)
Wealth dec.	0.017*** (0.001)	0.025*** (0.001)	0.016*** (0.001)	0.0003*** (0.0001)	0.005*** (0.001)	0.192*** (0.004)
Income dec.	0.009*** (0.001)	0.020*** (0.001)	0.008*** (0.001)	-0.000 (0.000)	0.008*** (0.001)	0.186*** (0.005)
Home Owner	-0.057*** (0.003)	-0.074*** (0.004)	-0.049*** (0.003)	-0.001*** (0.000)	-0.007*** (0.004)	-0.469*** (0.021)
Hours Worked	-0.000 (0.000)	-0.001*** (0.000)	-0.000 (0.000)	0.000 (0.000)	0.0003*** (0.0001)	-0.004*** (0.001)
Other Controls	Yes	Yes	Yes	Yes	Yes	Yes
Region#Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Occupation M and F FE	Yes	Yes	Yes	Yes	Yes	Yes

(a) Table 5: asset allocation and diversification

Table 6
Spousal equality and financial returns.
This table shows the coefficient estimates for a linear regression of the ratio of financial income over financial assets (columns (1) and (4)), the ratio of capital income over total assets (columns (2) and (5)), and the ratio of net capital income over total net assets (columns (3) and (6)). The sample consists of all two-spouse households in columns (1) to (3), and panel households in columns (3) to (6). In columns (3) to (6) the outcome variable is the household average across all survey years. Standard errors are bootstrapped in columns (1) to (3) and clustered at the region level in columns (3) to (6). *, **, and *** indicate statistical significance at the 10%, 5%, and 1%, respectively.

	Full sample			Panel sample		
	Financial return	Total return	Total net return	Financial return	Total return	Total net return
Equality	0.319*** (0.086)	0.431*** (0.072)	0.632*** (0.129)	0.706*** (0.148)	0.576*** (0.173)	0.806*** (0.354)
Other Controls	Yes	Yes	Yes	Yes	Yes	Yes
Region#Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Occupation M and F FE	Yes	Yes	Yes	Yes	Yes	Yes
Sector HH FE	Yes	Yes	Yes	Yes	Yes	Yes
adj. R ²	0.716	0.371	0.249	0.665	0.359	0.254
Observations	54,775	62,604	62,747	10,832	11,560	11,571

(b) Table 6: financial and total returns

Economic message: Equality changes not only whether households participate, but also how they allocate assets and how well they perform. This is the main evidence that patriarchal norms carry economic costs.

7.8 Table 7: Role of collaboration; scope and incentives

Read:

- Panel (a) tests whether Equality effects are stronger when collaboration should be more valuable.
- The effect is weaker when spouses work in the same sector, consistent with less information complementarity.
- The effect is stronger when the non-head spouse has more available time and when the couple has children.
- Panel (b) tests scope and incentives.
- Equality effects are weaker among couples with both very low or very high education, weaker among foreign-born couples, and decreasing in wealth.

Table 7

Spousal equality and investments.

This table shows the coefficient estimates for a linear regression of an indicator variable that takes the value 1 if the household holds wealth in at least one financial asset other than bank deposits.

(a) The Role of Collaboration

Spouse: Fin. Sector is an indicator variable that takes value 1 if the non-head spouse is employed in the financial sector. *Spouse: Same Sector* is an indicator variable that takes value 1 if the non-head spouse is employed in the same sector as the household head. *Hours Worked Ratio* is the ratio of hours worked by the household head over hours worked by the non head spouse. *Couple with Children* is an indicator variable that takes value 1 if there are children in the household. *X Var* indicates interaction terms of *Var* with *Equality*. Standard errors (in parentheses) are computed with a 2-stages bootstrapping procedure for generated variables. *, **, and *** indicate statistical significance at the 10%, 5%, and 1%, respectively.

	(1)	(2)	(3)	(4)
Equality	0.124*** (0.027)	0.136*** (0.027)	0.160*** (0.029)	0.084*** (0.030)
X Spouse: Fin. Sector	0.054 (0.109)			
X Spouse: Same Sector		-0.054* (0.027)		
X Hours Worked Ratio			-0.012* (0.007)	
X Couple with Children				0.047*** (0.016)
Other Controls	Yes	Yes	Yes	Yes
Region#Year FE	Yes	Yes	Yes	Yes
Occupation M and F FE	Yes	Yes	Yes	Yes
Sector HH FE	Yes	Yes	Yes	Yes

(a) Table 7(a): role of collaboration

adj. R ²	0.276	0.276	0.273	0.276
Observations	63,457	63,457	34,581	63,457

(b) Scope and Incentives

In column (1), we restrict the sample to households where both spouses have education equal or below 2 (primary school) and equal or above 4 (college or above). In column (2), we restrict the sample to households where one or both spouses are foreign born. In column (3), we pool the samples in column (1) and (2). In column (4) we exclude observations belonging to the subsample in column (3). In column (5) we include all couple-households. *Equality X Wealth Dec.* indicates interaction terms of *Equality* with *Wealthdec*. Standard errors (in parentheses) are clustered at the region level in columns (1), (2), (3), and (5). Bootstrapped standard errors (in parentheses) in column (4) are computed using random draws of size 21,205 from the full sample excluding observations belonging to the subsamples in columns (1) and (2). *, **, and *** indicate statistical significance at the 10%, 5%, and 1%, respectively.

	(1) Low/high education	(2) Foreign born	(3) Bootstrapped full sample	(4) Equality and wealth
Equality	0.073 (0.048)	0.028 (0.098)	0.156*** (0.046)	0.207*** (0.042)
Equality X Wealth Dec.				-0.013*** (0.004)
Other Controls	Yes	Yes	Yes	Yes
Region#Year FE	Yes	Yes	Yes	Yes
Occupation HH FE	Yes	Yes	Yes	Yes
Sector HH FE	Yes	Yes	Yes	Yes
N	18,080	3617	42,667	63,457
adj. R ²	0.280	0.268	0.267	0.276

(b) Table 7(b): scope and incentives

Economic message: The mechanism evidence fits the model. Equality matters most when spousal collaboration and correct head selection are likely to improve decision quality, and less when norms are less relevant or less costly to ignore.

7.9 Table 8: Non-couple households

Read:

- Table 8 is the main placebo test.
- Panel (a) shows that Equality does not significantly predict non-couple households' participation in financial assets, stocks, bonds, or other securities.
- Panel (b) shows no significant effect on financial returns, total returns, or total net returns for male- or female-headed non-couple households.
- Female headship itself is negatively associated with some participation outcomes among non-couple households, but Equality is not.

Table 8

Non-couple households.

(a) Investments

Column (1) shows the coefficient estimates for a linear regression of an indicator variable that takes value 1 if the household holds wealth in at least one financial asset other than bank deposits. Columns (2), (3), and (4) show coefficient estimates for a linear regression of an indicator variable that takes value 1 if the household holds wealth in stocks, bonds (either government or corporate), and other securities, respectively. FEs include *Region#Year*, *Occupation*, and *Sector* fixed effects. The sample consists of all non-couple households. Standard errors (in brackets) are clustered at the region level. *, **, and *** indicate statistical significance at the 10%, 5%, and 1%, respectively.

	(1) Any fin. asset	(2) Stocks	(3) Bonds	(4) Other securities
Equality	0.037 (0.036)	0.038 (0.025)	-0.001 (0.028)	-0.001 (0.002)
Female Head	-0.028*** (0.004)	-0.027*** (0.005)	-0.014*** (0.005)	-0.000** (0.000)
Education	0.022*** (0.006)	0.006*** (0.002)	0.014** (0.006)	0.000 (0.000)
Age	0.007*** (0.001)	0.002** (0.001)	0.004*** (0.001)	0.000 (0.000)
Age ²	-0.000*** (0.000)	-0.000*** (0.000)	-0.000*** (0.000)	-0.000 (0.000)
Wealth dec.	0.044*** (0.005)	0.012*** (0.003)	0.033*** (0.004)	0.000 (0.000)
Income dec.	0.035*** (0.003)	0.008*** (0.001)	0.028*** (0.003)	0.000 (0.000)
Home Owner	-0.120*** (0.015)	-0.040*** (0.009)	-0.090*** (0.010)	-0.000 (0.000)
Other Controls	Yes	Yes	Yes	Yes
FEs	Yes	Yes	Yes	Yes
N	31,485	31,485	31,485	31,485
adj. R ²	0.265	0.094	0.194	0.005

(a) Table 8(a): non-couple investment outcomes

(b) Financial Returns

This table shows the coefficient estimates for a linear regression of the ratio of financial income over financial assets (columns (1) and (2)), the ratio of capital income over total assets (columns (3) and (4)), and the ratio of net capital income over total net assets (columns (5) and (6)). FEs include *Region#Year*, *Occupation*, and *Sector* fixed effects. The sample consists of all non-couple households. Standard errors (in parentheses) are clustered at the cohort-region level. *, **, and *** indicate statistical significance at the 10%, 5%, and 1%, respectively.

	(1) Financial return (male head)	(2) Financial return (female head)	(3) Total return (male head)	(4) Total return (female head)	(5) Total net return (male head)	(6) Total net return (female head)
Equality	-0.149 (0.193)	-0.291 (0.185)	0.0168 (0.174)	-0.119 (0.160)	0.194 (0.345)	-0.308 (0.227)
Other Controls	Yes	Yes	Yes	Yes	Yes	Yes
FEs	Yes	Yes	Yes	Yes	Yes	Yes
N	8070	16,136	9460	20,787	9522	20,865
adj. R ²	0.744	0.747	0.402	0.394	0.280	0.287

ally, non-couple households have lower in-
alth. Importantly, however, the coefficients for
ge, education, income and wealth are remark-

sults suggest that *Equality* does not relate to ir-
through its possible correlation with (omitted)
of female labor market development (see Table

(b) Table 8(b): non-couple return outcomes

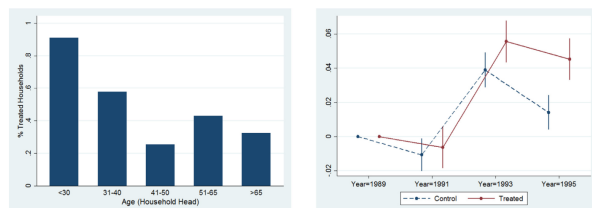
Economic message: The placebo results strongly support the intra-household mechanism. If Equality were merely regional modernization, trust, secularization, or female labor-market opportunity, it should also predict financial behavior among singles.

7.10 Figure 8: Pension reform; Table 9: Treated versus control households

Read:

- Figure 8 describes exposure to the 1992 pension reform and the female-headship trend for treated and control households.
- Treated households are not only the youngest; some middle-aged households are treated because they include younger employed members.
- The right-hand panel shows that female headship rises more among treated households after the reform.
- Table 9 shows that treated households are younger, higher income, have more working members, and have a higher female-to-male income ratio.

least one household member affected by the 1992 pension reform. The figure on the right hand side shows year fixed effects for two regression controls include average education, income decile, female-to-male income ratio and region fixed effects.



(a) Figure 8: pension reform treatment and trends

Table 9

Pension reform: treated vs. control households.

This table shows mean values and standard deviations for selected characteristics of treated and control households. Treated households have at least one household member affected by the 1992 pension reform. The sample consists of two-spouse households in the 1989, 1991, 1993, and 1995 surveys.

	Control	Treated	Total
Education (Couple Avg.)	2.930 (0.935)	3.101 (0.981)	3.003 (0.959)
Income Decile	5.941 (2.641)	7.153 (2.363)	6.461 (2.596)
# Children	1.520	1.571	1.541

(b) Table 9: treated versus control households

Economic message: The pension reform created heterogeneous exposure to a shock that raised the need for private financial planning. The descriptive evidence is consistent with the reform weakening patriarchal headship norms.

7.11 Table 10: Effects of the pension reform

Read:

- Table 10 estimates difference-in-differences effects of pension reform exposure.
- Treated households become 2.1 percentage points more likely to have female headship after the reform.

- Treated households increase saving rates by about 8.7 percentage points.
- They also increase pension fund participation, female hours worked, and the female-to-male income ratio.
- Male hours worked increases only modestly and insignificantly relative to female hours worked.

Table 10

Effects of the pension reform.

This table shows the coefficient estimates for a linear regression of female headship (column (1)), savings (column (2)), investment in pension funds (column (3)), hours worked by female spouse (column (4)), hours worked by male spouse (column (5)), and ratio of female spouse income over male spouse income (column (6)). *Post* is an indicator variable that takes the value 1 after the pension reform, that is, in years 1993 and 1995. *Treated* is an indicator variable that takes value 1 if at least one household member is affected by the reform. The sample consists of two-spouse households in the 1989, 1991, 1993, and 1995 surveys. Robust standard errors in parentheses. *, **, and *** indicate statistical significance at the 10%, 5%, and 1%, respectively.

	(1) Female headship	(2) Saving rate	(3) Pension funds	(4) Hours worked (F)	(5) Hours worked (M)	(6) Income ratio FtoM
Post X Treated	0.021*** (0.006)	0.087*** (0.024)	0.025*** (0.008)	1.161*** (0.441)	0.357 (0.304)	0.029** (0.011)
Post	0.026*** (0.004)	-0.090*** (0.013)	0.015*** (0.005)	-0.782*** (0.297)	0.109 (0.203)	0.000 (0.007)
Treated	-0.023*** (0.003)	0.079*** (0.012)	-0.017*** (0.005)	0.930*** (0.334)	-2.271*** (0.209)	0.004 (0.008)
Education (Couple Avg.)	-0.005*** (0.002)	-0.092*** (0.008)	0.011*** (0.002)	1.058*** (0.144)	-0.370*** (0.102)	0.033*** (0.004)
Income Decile	0.002** (0.001)	0.117*** (0.004)	0.011*** (0.001)	2.211*** (0.055)	0.592*** (0.042)	0.037*** (0.001)
% Children	-0.072*** (0.015)	-0.317*** (0.063)	0.033*** (0.013)	-8.604*** (0.832)	8.302*** (0.685)	-0.413*** (0.026)
% HH Members in LF	-0.115*** (0.026)	-0.406*** (0.124)	0.098*** (0.025)	4.443*** (1.709)	17.021*** (1.519)	-0.299*** (0.050)
Cohort (HH Head) FE	Yes	Yes	Yes	Yes	Yes	Yes
Sector (HH Head) FE	Yes	Yes	Yes	Yes	Yes	Yes
Region FE	Yes	Yes	Yes	Yes	Yes	Yes
N	20,312	20,243	20,312	20,165	20,264	20,185
adj. R ²	0.088	0.138	0.056	0.232	0.657	0.128

Table 10: pension reform effects

Economic message: The reform did not only change saving. It also changed household governance. When the economic cost of inefficient financial management rose, families became more willing to move away from male-default headship.

8 Short summary

- The paper studies how gender norms shape household financial decisions in Italy.
- Female household financial headship increased sharply from the early 1990s to 2014.
- The authors build a social-conformism model in which households trade off efficient allocation of financial management against the desire to conform to gender roles.
- They estimate gender norms using cohort-region fixed effects in female-headship regressions after controlling for spousal comparative advantage.
- The resulting Equality measure predicts financial-market participation, holdings of risky and safe financial assets, pension fund participation, diversification, and returns.
- Mechanism tests support both selection and collaboration channels.
- Placebo tests on non-couple households show no comparable effects, reducing concerns about broad omitted variables.
- The 1992 Italian pension reform increased female headship among treated households, suggesting that economic shocks can accelerate cultural change.

- The core takeaway is that patriarchal norms can distort household financial management and reduce household welfare.

9 Conclusion

9.1 Core conclusion

Guiso and Zaccaria show that gender norms inside households are economically consequential. More egalitarian social norms increase financial-market participation, broaden asset allocation, improve diversification, and raise measured returns. The evidence suggests that excluding women from household financial decisions because of traditional gender roles can reduce household welfare.

9.2 Economic intuition

The intuition is simple. Financial management is a task that requires attention, information, and ability. If a patriarchal norm assigns the task to the husband by default, the household may ignore the spouse with better skills or better information. Egalitarian norms relax this constraint. They allow the household to choose the better decision maker or to make decisions more collaboratively. This improves participation and portfolio performance.

9.3 Incremental contribution revisited

The paper contributes by measuring gender norms from observed family governance rather than survey attitudes, linking those norms to household finance outcomes, and showing how an economic policy shock can accelerate norm change. It bridges household finance, cultural economics, gender economics, and intra-household allocation.

9.4 Implications for household finance

The findings imply that household financial mistakes may not be only the result of low literacy, participation costs, or risk aversion. They may also arise from social rules that decide whose knowledge and attention matter. Improving financial outcomes may therefore require changing intra-household decision processes, not only providing information or lowering transaction costs.

9.5 Implications for gender economics

The paper shows that gender equality has benefits beyond labor-market outcomes. Even if women do not earn more than men or work in finance, allowing them to participate in household financial management can improve household portfolio choices and returns.

9.6 Limitations

The baseline design is observational, and Equality is a generated measure based on survey-reported headship. Some omitted cohort-region factors could in principle remain. Return measures are

approximate because SHIW is not a transaction-level portfolio dataset. The Italian context may also be special because of the importance of family governance, public pensions, and regional cultural heterogeneity.

9.7 Takeaway

The enduring takeaway is that social norms can act like hidden constraints in household finance. When the norm says that men should manage money, households may make worse financial decisions. When norms become more egalitarian, financial decision making becomes more efficient, diversified, and better performing.